

DevelopersHub Corporation

CI/CD QA Workflow Document

Week 6 - QA Integration into DevOps Pipeline

GitHub Repository: <https://github.com/kashan-X/SQA-Internship-DevelopersHub>

Tools: GitHub Actions, Selenium, JMeter, OWASP ZAP

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1. Introduction to CI/CD for QA

Continuous Integration / Continuous Delivery (CI/CD) is a DevOps practice where code changes are automatically built, tested, and deployed. For QA engineers, CI/CD means automated tests run on every code push, giving instant feedback on quality without manual intervention.

Term	Definition	QA Role
Continuous Integration (CI)	Developers merge code frequently into a shared repo	Automated tests run on every merge
Continuous Delivery (CD)	Code is always in a deployable state	QA validates before release
Pipeline	A series of automated steps from code to deployment	QA tests are steps in the pipeline
Trigger	Event that starts the pipeline (e.g. push, pull request)	Tests auto-run on code changes
Test Report	Output of automated test runs	QA reviews reports for failures
Artifact	Output files from pipeline (test results, reports)	JMeter/ZAP reports saved as artifacts

2. Role of QA in CI/CD Pipeline

QA engineers are responsible for integrating automated tests at multiple stages of the CI/CD pipeline. This ensures that every code change is validated before it reaches production.

Pipeline Stage	QA Activity	Tools Used
Code Push / PR	Run unit & integration tests	JUnit, PyTest, Selenium
Build Stage	Verify build succeeds with all dependencies	Maven, npm, pip
Test Stage	Run automated functional & regression tests	Selenium WebDriver
Performance Stage	Run load/stress tests on staging environment	Apache JMeter
Security Stage	Run automated security scans	OWASP ZAP
Deploy Stage	Smoke tests after deployment	Postman, Selenium
Production	Monitoring & alerting	New Relic, Datadog

3. GitHub Actions Overview

GitHub Actions is a CI/CD platform built directly into GitHub. It allows QA engineers to define workflows in YAML files that automatically trigger when code is pushed, pull requests are opened, or on a schedule.

Feature	Description
Workflow File	.yml file stored in .github/workflows/ folder
Trigger (on:)	Defines when the workflow runs (push, pull_request, schedule)
Jobs	Groups of steps that run on a virtual machine (ubuntu, windows, mac)
Steps	Individual commands or actions within a job
Actions	Pre-built reusable components (e.g. actions/checkout, actions/setup-java)
Artifacts	Files saved after workflow run (test reports, screenshots)
Secrets	Encrypted environment variables (passwords, API keys)

4. Proposed CI/CD Workflow for automationexercise.com QA

Below is the complete CI/CD pipeline designed to automate QA testing for automationexercise.com. This workflow would run on every code push to the main branch.

Step	Stage	Action	Tool	Output
1	Code Push	Developer pushes code to GitHub	Git / GitHub	Triggers pipeline

2	Checkout	Clone repository to test runner	actions/checkout	Code available
3	Setup Java	Install Java 11 for JMeter	actions/setup-java	Java ready
4	Install Dependencies	Install Python, Selenium, ChromeDriver	pip install	Test env ready
5	Functional Tests	Run Selenium test scripts on staging URL	Selenium + Chrome	Pass/Fail report
6	Performance Test	Run JMeter load test (50 users)	Apache JMeter	JMeter HTML report
7	Security Scan	Run OWASP ZAP baseline scan	OWASP ZAP Action	ZAP alerts report
8	Publish Reports	Upload all reports as pipeline artifacts	actions/upload-artifact	Reports saved
9	Notify Team	Send email/Slack notification of results	Slack GitHub Action	Team notified
10	Deploy (CD)	Deploy to production if all tests pass	GitHub Pages / Heroku	Live deployment

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7.

Trigger	When it Fires	Tests That Run
push to main	Every time code is merged to main branch	All 3 jobs (Selenium + JMeter + ZAP)
pull_request	When a PR is opened or updated	Selenium functional tests only
schedule (cron)	Every Monday at 9:00 AM automatically	Full pipeline (all 3 jobs)
Manual (workflow_dispatch)	QA engineer clicks 'Run workflow' in GitHub	Selected jobs only

8. Test Report Generation in Pipeline

Report Type	Generated By	Format	Where Saved
Functional Test Report	Selenium + pytest-html	HTML	GitHub Artifacts
Performance Test Report	JMeter -e flag	HTML Dashboard	GitHub Artifacts
Security Scan Report	OWASP ZAP Action	HTML	GitHub Artifacts
Pipeline Summary	GitHub Actions built-in	Web UI	GitHub Actions tab

9. GitHub Repository

All test scripts, reports, and documentation are uploaded to:

Repository URL: <https://github.com/kashan-X/SQA-Internship-DevelopersHub>

Repository Structure:

